

Course intro and containerization

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- Course git repo (coming soon): slides, homework specs, starters
- Textbook: Poldrack, R., **Better Code, Better Science**

- Docker, shell, SSH

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- pytest, TDD, validation
- Other lab tools

- Introduction to containerization

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- Start the class Docker container

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- Start the class Docker container
- Connect with VSCode

The goal of containerization is to ensure consistent and portable computational environments. That is, its goal is to create a computational system that reliably works the same on every computer. This is especially useful in computational research, where analyses can be complex and depend on a lot of interacting pieces of software and code that may come in many versions.

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We will use Docker Desktop to understand the basic ideas and practical applications.

What is containerization?

Containerization is a way to package software and everything it needs to run into a single “box,” called a **container**. This box can be moved to any computer, and the software will run exactly the same way. Containerization is a lot like virtualization but tends to be more efficient.

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Below, we will also talk about the **image**, which something like a template to make a kind of container. We can have many images, and each image can spawn many containers. Often, it's the image we will share to others.

When you buy a piece of furniture from IKEA, the package includes everything you need: screws, tools, instructions, and the wooden pieces. Even if you don't have your own tools, you can still assemble the furniture because everything is in the box. Similarly, the **image** provides all the “tools” and “pieces” a program needs to run, so it works the same way on any computer. The **container** is like the finished piece of furniture.

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- **Consistency:** Containers ensure the same program works the same way, no matter whose computer it's on.
- **Portability:** You can move containers between your laptop, a friend's computer, or even powerful online servers.
- **Simplicity:** Containers bundle everything needed, so you don't have to install things one by one.

1. Images

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- A **container** is a running version of an image.
- Think of it as the assembled IKEA furniture created from the box's contents.

3. Docker and Docker Desktop

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Docker Desktop is the specific application we will use to work with Docker containers using a graphical interface and minimal technical setup.

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- Open the application, and follow the setup instructions.
- Verify the installation by running the “Hello World” container (from Docker Desktop).

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- Port mapping assigns a port on the **host** computer to a port on the **container** so you can use its services.

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- **Graphical interface:** No need to memorize commands; most operations are clickable.
- **Cross-platform:** Works on Windows, Mac, and Linux.
- **Prebuilt containers:** Access ready-to-use containers for popular programs with just a few clicks.

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- Programs often need specific versions of tools or software to work.
- What works on one computer might not work on another.
- Sharing programs can be tricky because everyone's computer is set up differently.
- Changing your local setup for one program might break others.

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- **Encapsulation:** Everything the program needs is packaged together.
- **Portability:** Containers run the same way everywhere.
- **Reproducibility:** You can guarantee the same results, no matter where or when the program is run.

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- **For teams:** Everyone works with the same tools, avoiding conflicts.
- **For sharing:** You can send a image to someone, and it will just work.
- **For researchers:** Reviewers and readers can download your Docker image and repeat your analyses.

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- Docker Desktop makes creating and using containers simple and efficient.
- Containers solve problems of consistency, portability, and reproducibility.

- Download and install Docker Desktop on your computer.

Next steps

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- Run the “Hello World” container to verify your setup.

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- Go to the next chapter